Semester	B.E. Semester VII – CMPN
Subject	R Programming LAB
Subject Professor In-charge	Prof. Sanjeev Dwivedi
Assisting Teachers	
Laboratory	M411

Student Name	Eshanika Amballa	
Roll Number	23102B0056	
Grade and Subject Teacher's Signature		

Experiment Number	05
Experiment Title	Social Network Analysis with R
Objectives (Skill Set / Knowledge Tested / Imparted)	You are required to study the Social Network Analysis with R project demonstrated in the prescribed video tutorial and implement the demonstrated network-analysis workflow using R Programming.
Description	You are required to study the Social Network Analysis with R project demonstrated in the prescribed video tutorial and implement the demonstrated network-analysis workflow using R Programming. The project should focus on representing relationships among entities as a network, analyzing the structure of the network, identifying important or influential nodes, and visualizing the

	resulting social network using suitable R packages and functions. After completing the implementation, execute and test the analysis, interpret the generated network measures and visualizations, and systematically organize the R scripts, dataset, and generated outputs. Reference Video Social Network Analysis with R https://www.youtube.com/watch?v=0xsM0MbRPGE&list=PLpaxqU9O4beceS4RxsJESmFGUqpw-m2_P&index=1&t=871s The supplied video is also referenced as learning material for R-based social/network analytics in academic course material.

Program	# load required package
	library(igraph)
	
	g <- graph(c(1,2, 2,3, 3,4, 4,1),
	        directed = F,
	        n = 7)
	
	plot(g,
	    vertex.color = "steelblue",
	    vertex.size = 25,
	    main = "Simple Undirected Graph (7 Nodes)")
	
	g1 <- graph(c("Amy", "Ram", "Ram", "Li", "Li", "Amy",
	          "Amy", "Li", "Kate", "Li"),
	         directed = T)

```
plot(g1,
     vertex.color = "orange",
     vertex.size = 30,
     edge.arrow.size = 0.5,
     main = "Directed Graph – Named Nodes")


# Degree centrality (all, in, out)
cat("===== Degree Centrality =====\n")
cat("All:\n")
print(degree(g1, mode = 'all'))

cat("\nIn-degree:\n")
print(degree(g1, mode = 'in'))

cat("\nOut-degree:\n")
print(degree(g1, mode = 'out'))

# Diameter
cat("\nDiameter:", diameter(g1, directed = F, weights = NA), "\n")

# Edge density
cat("Edge Density:", edge_density(g1, loops = F), "\n")

# Manual edge density calculation
cat("Manual Edge Density:", ecount(g1) / (vcount(g1) *
(vcount(g1) - 1)), "\n")

# Reciprocity
cat("Reciprocity:", reciprocity(g1), "\n")

# Closeness centrality
cat("\n===== Closeness Centrality =====\n")
print(closeness(g1, mode = 'all', weights = NA))

# Betweenness centrality
cat("\n===== Betweenness Centrality =====\n")
print(betweenness(g1, directed = T, weights = NA))
```

```
# Edge betweenness
cat("\n===== Edge Betweenness =====\n")
print(edge_betweenness(g1, directed = T, weights = NA))

data <- read.csv('https://raw.githubusercontent.com/bkrai/R-files-from-YouTube/main/networkdata.csv',
             header = T)

cat("\n===== Dataset Preview =====\n")
print(head(data))
cat("Total rows:", nrow(data), "\n")

y <- data.frame(data$first, data$second)


net <- graph.data.frame(y, directed = T)
V(net)$label <- V(net)$name
V(net)$degree <- degree(net)

hist(V(net)$degree,
    col = "steelblue",
    main = "Histogram of Node Degree",
    xlab = "Degree",
    ylab = "Frequency")


plot(net,
    main = "Network Diagram – Basic")


plot(net,
    vertex.color = rainbow(52),
    vertex.size  = V(net)$degree * 0.4,
    edge.arrow.size = 0.1,
    layout = layout.fruchterman.reingold,
    main   = "Network – Nodes Sized by Degree (Fruchterman-Reingold)")
```

```
hs <- hub_score(net)$vector
as <- authority.score(net)$vector

par(mfrow = c(1, 2))

set.seed(123)

plot(net,
    vertex.size  = hs * 30,
    main = 'Hubs',
    vertex.color = rainbow(52),
    edge.arrow.size = 0.1,
    layout = layout.kamada.kawai)

plot(net,
    vertex.size  = as * 30,
    main = 'Authorities',
    vertex.color = rainbow(52),
    edge.arrow.size = 0.1,
    layout = layout.kamada.kawai)

par(mfrow = c(1, 1))


net <- graph.data.frame(y, directed = F)
cnet <- cluster_edge_betweenness(net)

plot(cnet, net,
    main = "Community Detection (Edge Betweenness)")

cat("\n===== Community Detection Results =====\n")
cat("Number of communities:", length(cnet), "\n")
cat("Modularity:", round(modularity(cnet), 4), "\n")
cat("\nCommunity memberships:\n")
print(membership(cnet))
```

	cat("1. The small directed graph (Amy, Ram, Li, Kate) demonstrates\n") cat(" basic network properties — degree, closeness, betweenness,\n") cat(" reciprocity, and edge density.\n\n") cat("2. The larger network from networkdata.csv contains 52 nodes\n") cat(" with directed relationships. The degree histogram shows\n") cat(" how connections are distributed across nodes.\n\n") cat("3. Hub scores identify nodes that point to many important\n") cat(" authorities. Authority scores identify nodes that are\n") cat(" pointed to by many important hubs.\n\n") cat("4. Community detection using edge betweenness clustering\n") cat(" reveals natural groupings within the undirected version\n") cat(" of the network — nodes within the same community are\n") cat(" more densely connected to each other than to outsiders.\n\n") cat("==\n") cat(" Analysis Complete. \n") cat("==\n")
Output	

```
> g <- graph(c(1,2, 2,3, 3,4, 4,1),
+             directed = F,
+             n = 7)

> plot(g,
+       vertex.color = "steelblue",
+       vertex.size = 25,
+       main = "Simple Undirected Graph (7 Nodes)")

> g1 <- graph(c("Amy", "Ram", "Ram", "Li", "Li", "Amy",
+                "Amy", "Li", "Kate", "Li"),
+              directed = T)

> plot(g1,
+       vertex.color = "orange",
+       vertex.size = 30,
+       edge.arrow.size = 0.5,
+       main = "Directed Graph – Named Nodes")

> # Degree centrality (all, in, out)
> cat("===== Degree Centrality =====\n")
===== Degree Centrality =====

> cat("All:\n")
All:

> print(degree(g1, mode = 'all'))
 Amy  Ram   Li Kate 
   3    2    4    1 

> cat("\nIn-degree:\n")

In-degree:
```

```
Console  Terminal  Background Jobs
R 4.6.1 · ~/
> print(degree(g1, mode = 'in'))
 Amy  Ram   Li Kate 
   1    1    3    0 

> cat("\nOut-degree:\n")

Out-degree:

> print(degree(g1, mode = 'out'))
 Amy  Ram   Li Kate 
   2    1    1    1 

> # Diameter
> cat("\nDiameter:", diameter(g1, directed = F, weights = NA), "\n")

Diameter: 2

> # Edge density
> cat("Edge Density:", edge_density(g1, loops = F), "\n")
Edge Density: 0.4166667

> # Manual edge density calculation
> cat("Manual Edge Density:", ecount(g1) / (vcount(g1) * (vcount(g1) - 1)), "\n")
Manual Edge Density: 0.4166667

> # Reciprocity
> cat("Reciprocity:", reciprocity(g1), "\n")
Reciprocity: 0.4

> # Closeness centrality
> cat("\n===== Closeness Centrality =====\n")
```

```
> print(closeness(g1, mode = 'all', weights = NA))
      Amy       Ram        Li      Kate 
0.2500000 0.2500000 0.3333333 0.2000000 

> # Betweenness centrality
> cat("\n===== Betweenness Centrality =====\n")

===== Betweenness Centrality =====

> print(betweenness(g1, directed = T, weights = NA))
 Amy  Ram   Li Kate 
   2    0    3    0 

> # Edge betweenness
> cat("\n===== Edge Betweenness =====\n")

===== Edge Betweenness =====

> print(edge_betweenness(g1, directed = T, weights = NA))
[1] 3 2 5 1 3

> data <- read.csv('https://raw.githubusercontent.com/bkrai/R-files-from-YouTube/main/networkdata.csv',
+                  header = T)

> cat("\n===== Dataset Preview =====\n")

===== Dataset Preview =====

> print(head(data))
  first second grade spec
1    AA     DD     6    Y
2    AB     DD     6    R
3    AF     BA     6    Q
4    DD     DA     6    Q
5    CD     EC     6    X
6    DD     CE     6    Y
```

Console | Terminal | Background Jobs

R 4.6.1 · ~/

```
> cat("Total rows:", nrow(data), "\n")
Total rows: 290 

> y <- data.frame(data$first, data$second)

> net <- graph.data.frame(y, directed = T)

> V(net)$label <- V(net)$name

> V(net)$degree <- degree(net)

> hist(V(net)$degree,
+      col = "steelblue",
+      main = "Histogram of Node Degree",
+      xlab = "Degree",
+      ylab = "Frequency")

> plot(net,
+      main = "Network Diagram - Basic")

> plot(net,
+      vertex.color = rainbow(52),
+      vertex.size  = V(net)$degree * 0.4,
+      edge.arrow.size = 0.1,
+      layout = layout.fruchte .... [TRUNCATED]

> hs <- hub_score(net)$vector

> as <- authority.score(net)$vector

> par(mfrow = c(1, 2))

> set.seed(123)
```

Source

Console Terminal Background Jobs

R 4.6.1 · ~/

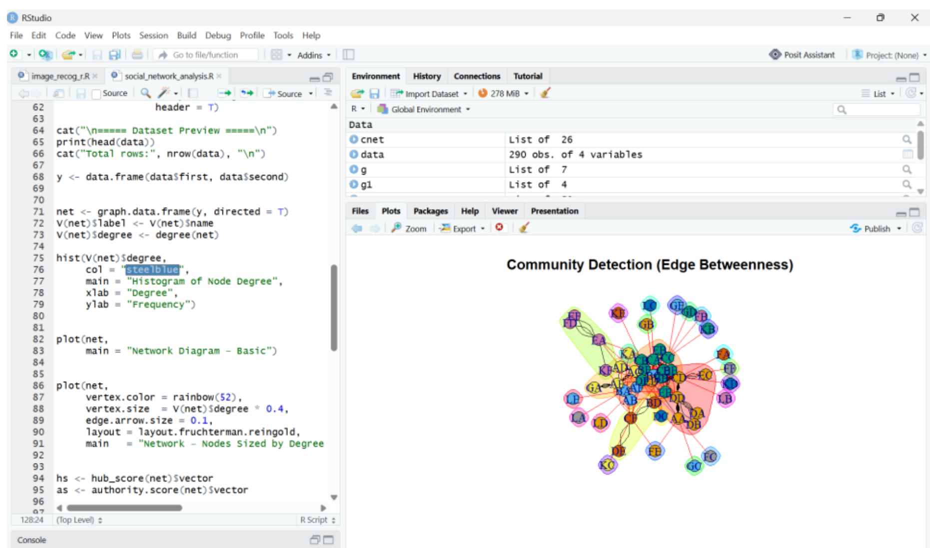
```
> plot(net,
+      vertex.size  = hs * 30,
+      main = 'Hubs',
+      vertex.color = rainbow(52),
+      edge.arrow.size = 0.1,
+      layout = layo .... [TRUNCATED]

> plot(net,
+      vertex.size  = as * 30,
+      main = 'Authorities',
+      vertex.color = rainbow(52),
+      edge.arrow.size = 0.1,
+      layout .... [TRUNCATED]

> par(mfrow = c(1, 1))

> net <- graph.data.frame(y, directed = F)

> cnet <- cluster_edge_betweenness(net)

> plot(cnet, net,
+      main = "Community Detection (Edge Betweenness)")

> cat("\n===== Community Detection Results =====\n")

===== Community Detection Results =====

> cat("Number of communities:", length(cnet), "\n")
Number of communities: 26

> cat("Modularity:", round(modularity(cnet), 4), "\n")
Modularity: 0.3514

> cat("\nCommunity memberships:\n")
```

Console Terminal Background Jobs

R 4.6.1 · ~/

```
Community memberships:

> print(membership(cnet))
AA AB AF DD CD BA CB CC BC ED AE CA EB BF BB AC DC BD DB CF DF BE EA CE EE EF FF FD
 1  2  2  1  1  2  3  3  2  1  4  3  3  3  3  4  5  6  1  6  3  3  7  3  3  7  8  7
GB GC GD AD KA KF LC DA EC FA FB DE FC FE GA GE KB KC KD KE LB LA LD LE
 9 10 11  4 12  7 13  1  1 14 15  6 16 17  4 18 19 20 21 22 23 24 25 26

> cat("1. The small directed graph (Amy, Ram, Li, Kate) demonstrates\n")
1. The small directed graph (Amy, Ram, Li, Kate) demonstrates

> cat("   basic network properties - degree, closeness, betweenness,\n")
   basic network properties - degree, closeness, betweenness,

> cat("   reciprocity, and edge density.\n\n")
   reciprocity, and edge density.


> cat("2. The larger network from networkdata.csv contains 52 nodes\n")
2. The larger network from networkdata.csv contains 52 nodes

> cat("   with directed relationships. The degree histogram shows\n")
   with directed relationships. The degree histogram shows

> cat("   how connections are distributed across nodes.\n\n")
   how connections are distributed across nodes.


> cat("3. Hub scores identify nodes that point to many important\n")
3. Hub scores identify nodes that point to many important

> cat("   authorities. Authority scores identify nodes that are\n")
   authorities. Authority scores identify nodes that are

> cat("   pointed to by many important hubs.\n\n")
   pointed to by many important hubs.
```

```
> cat("   how connections are distributed across nodes.\n\n")
   how connections are distributed across nodes.

> cat("3. Hub scores identify nodes that point to many important\n")
3. Hub scores identify nodes that point to many important

> cat("   authorities. Authority scores identify nodes that are\n")
   authorities. Authority scores identify nodes that are

> cat("   pointed to by many important hubs.\n\n")
   pointed to by many important hubs.

> cat("4. Community detection using edge betweenness clustering\n")
4. Community detection using edge betweenness clustering

> cat("   reveals natural groupings within the undirected version\n")
   reveals natural groupings within the undirected version

> cat("   of the network – nodes within the same community are\n")
   of the network – nodes within the same community are

> cat("   more densely connected to each other than to outsiders.\n\n")
   more densely connected to each other than to outsiders.

> cat("============================================================\n")
============================================================

> cat("             Analysis Complete.                             \n")
             Analysis Complete.

> cat("============================================================\n")
============================================================
```



Conclusion	This lab successfully demonstrated how R and the igraph package can be used to construct, visualize, and analyze social networks by computing centrality measures, identifying influential nodes through hub/authority scores, and detecting communities using edge betweenness clustering.